Operation Manual

PLC-Based Auto Labelling System (Conveyor Automation)

1. Introduction

1.1 Purpose of the Auto Labelling System

The auto labelling system is designed to automate the process of applying labels to products moving along a conveyor line. The system ensures consistent label placement, improved production efficiency, and reduced dependency on manual operations.

The primary objectives of the system are to:

- Automate product labelling within a continuous production flow
- Improve accuracy and repeatability of label placement
- Reduce manual intervention and operational delays
- Increase overall throughput in packaging operations

1.2 Overview of PLC-Based Automation

The system is controlled through a Programmable Logic Controller (PLC), which coordinates all inputs and outputs in real time. The PLC processes signals from sensors and executes predefined control logic to manage system operations.

Key automation functions include:

- Detection of products using sensors
- Sequential control of conveyor movement
- Timed activation of the labelling mechanism
- Coordination between field devices and the control interface

This PLC-based architecture ensures reliable, deterministic, and real-time control suitable for industrial automation environments.

1.3 Role of the System in the Production Line

The auto labelling system functions as an intermediate stage within a production or packaging line. It is typically positioned after product preparation processes and before final packaging or dispatch.

Its primary role includes:

- Detecting incoming products on the conveyor system
- Synchronizing conveyor movement with labelling actions
- Ensuring accurate and uniform label application
- Supporting continuous and efficient production flow

By integrating conveyor automation with PLC-based control logic, the system contributes to improved operational consistency and process efficiency.

2. System Overview

The auto labelling system is an integrated industrial automation solution designed to manage product handling, detection, and labelling operations on a conveyor-based production line. The system combines programmable control, sensing devices, actuators, and an operator interface to achieve coordinated and repeatable industrial operation.

The architecture is centered around a Programmable Logic Controller (PLC), which acts as the main control unit. The PLC continuously executes control logic in a cyclic scan process, reading input signals from field devices, processing the programmed logic, and updating output signals to control actuators. This real-time execution ensures deterministic and reliable system behavior suitable for industrial environments.

The conveyor subsystem is responsible for transporting products through different stages of the labelling process. It is driven by an electric motor controlled through PLC outputs, typically via motor starters or drive interfaces. The conveyor speed and operational state (start, stop, or hold) are managed by the PLC based on system conditions and sensor feedback. Proper synchronization of conveyor movement is critical to ensure accurate product positioning during labelling.

Product detection is achieved using industrial sensors such as photoelectric or proximity sensors installed along the conveyor path. These sensors continuously monitor product presence and position. When a product is detected, the sensor sends a digital input signal to the PLC. The PLC evaluates this signal in conjunction with timing logic to determine the precise moment for label application. This ensures that labels are applied consistently and in the correct position on each product.

The labelling mechanism is triggered by the PLC based on predefined control conditions. Once a valid detection signal is received and timing conditions are met, the PLC activates the labelling actuator. The synchronization between conveyor movement and labelling action is essential to avoid errors such as misaligned or missing labels.

An HMI (Human-Machine Interface) is integrated into the system to provide operator-level monitoring and control. The HMI displays real-time system status, including conveyor operation, sensor states, and alarm conditions. It also allows operators to switch between operating modes such as automatic, manual, and maintenance modes. In addition, the HMI provides control functions such as system start/stop, parameter adjustment, and fault reset. Communication between the HMI and PLC is established through an industrial communication protocol, enabling continuous data exchange and system supervision.

Overall, the system ensures coordinated interaction between mechanical, electrical, and control components, enabling efficient, accurate, and automated labelling operations within a continuous production environment.

3. Safety Instructions

The auto labelling system operates using electrical power, moving mechanical components, and automated control logic. Proper safety practices must be followed at all times to prevent injury, equipment damage, and operational failure. Only trained personnel should operate or service the system.

3.1 Electrical Safety Precautions

Before performing any inspection, maintenance, or wiring activities, the system power supply must be isolated and locked out where applicable. Users must ensure that all electrical panels remain closed during operation. Direct contact with live terminals, exposed wiring, or PLC input/output modules is strictly prohibited.

All electrical work should be carried out in accordance with applicable industrial electrical safety standards. Proper grounding and insulation must be maintained to reduce the risk of electric shock or equipment malfunction.

3.2 Emergency Stop Usage

The system is equipped with an emergency stop (E-Stop) function designed to immediately halt all machine operations in hazardous situations. When activated, the E-Stop disables conveyor movement, labelling actions, and all associated actuators.

The emergency stop should only be used in situations involving:

- Unexpected machine behavior
- Safety risks to personnel
- Mechanical failure or product jam conditions

After activation, the system must be reset only after the cause of the emergency condition has been identified and resolved.

3.3 Conveyor Pinch Point Hazards

The conveyor system contains moving belts, rollers, and drive mechanisms that create potential pinch point hazards. Operators must avoid placing hands, tools, or any objects near moving conveyor parts during operation.

Special caution must be taken at:

- Conveyor entry and exit points
- Roller joints and belt transitions

- Areas near drive motors and tension mechanisms

Physical guards and protective covers must remain in place during system operation.

3.4 Sensor and Moving Part Hazards

The system includes multiple sensors and moving actuators that operate automatically based on PLC logic. Sudden activation of conveyor movement or labelling units may occur without manual intervention when the system is in automatic mode.

Operators must not obstruct sensor paths or place objects in detection zones, as this may lead to unexpected system behavior. Additionally, care must be taken to avoid contact with moving labelling mechanisms during operation or system testing.

Proper awareness of system status indicators on the HMI is essential before performing any interaction with mechanical components.

4. System Components

The auto labelling system consists of several integrated hardware and control components that work together to achieve automated product handling, detection, and labelling on a conveyor line. Each component plays a specific role in ensuring coordinated and reliable system operation.

4.1 PLC Controller

The PLC acts as the central control unit of the system. It executes the programmed control logic, processes input signals from sensors, and sends output commands to actuators such as motors and the labelling unit. The PLC ensures real-time, deterministic control of all system operations.

4.2 Conveyor System (Product Transport Unit)

The conveyor system is responsible for continuous movement of products through the labelling station. It is driven by an electric motor controlled via the PLC. In configurations involving merging or multi-line transport, the conveyor ensures proper alignment and spacing of products for accurate processing.

4.3 Label Applicator Unit

The label applicator is the mechanical subsystem responsible for dispensing and applying labels onto products. It is activated by PLC output signals based on product detection and timing logic. The applicator ensures consistent placement and adhesion of labels during conveyor movement.

4.4 Sensors (Product Detection and Positioning)

Industrial sensors such as photoelectric or proximity sensors are used to detect the presence and position of products on the conveyor. These sensors provide real-time input signals to the PLC, enabling accurate synchronization of conveyor movement and labelling operations.

4.5 HMI Panel (Human-Machine Interface)

The HMI provides the operator interface for system monitoring and control. It displays real-time system status, operational modes, and alarm conditions. Operators can use the HMI to start or stop the system, switch between modes, and adjust operational parameters.

4.6 Motor Drives and Actuators

Motor drives control the speed and operation of conveyor motors based on PLC output signals. Additional actuators are used to control mechanical movements such as label dispensing. These components convert electrical control signals into physical motion required for system operation.

Together, these components form an integrated automation system in which mechanical movement, sensing, and control logic are synchronized through PLC-based control architecture to achieve efficient and accurate labelling operations.

5. Control System Overview

The control system is based on a Programmable Logic Controller (PLC) that manages the entire automation sequence of the auto labelling system. The PLC acts as the central processing unit, continuously scanning input signals, executing programmed control logic, and updating output signals in real time. This cyclic operation ensures precise coordination between mechanical and electrical components of the system.

The PLC controls the system using a structured input–process–output model. Input signals are received from field devices such as product detection sensors, limit switches, emergency stop circuits, and operator push buttons (start/stop/reset). These signals provide real-time information about system conditions, product presence, and operator commands. The PLC continuously monitors these inputs to determine the appropriate operational response based on the programmed logic.

Output signals from the PLC are used to control various actuators within the system. These include conveyor motor control signals, activation of the label applicator mechanism, and status indicators such as stack lights or alarm outputs. The PLC ensures that each output is triggered only when predefined logic conditions are satisfied, maintaining safe and synchronized operation of the system.

The HMI (Human-Machine Interface) is integrated with the PLC to provide a real-time monitoring and control platform for the operator. Communication between the HMI and PLC is established through an industrial communication protocol, enabling bidirectional data exchange. The HMI receives system status information from the PLC, such as operational mode, sensor states, and fault conditions, and displays them in a user-friendly format. At the same time, operator commands entered through the HMI, such as mode selection or system start/stop requests, are transmitted to the PLC for execution.

Overall, the control system ensures seamless coordination between input devices, control logic, and output actuators, enabling reliable and efficient automation of the labelling process.

6. Operating Modes

The auto labelling system is designed with multiple operating modes to support different stages of production, including normal operation, manual intervention, and system testing. Each mode controls the behavior of the conveyor, sensors, and labelling unit through PLC logic, ensuring safe and flexible system operation.

6.1 Automatic Mode (Production Mode)

In Automatic Mode, the system operates as a fully synchronized automated process without requiring continuous operator intervention. The PLC continuously monitors sensor inputs and executes the labelling sequence based on product detection.

In this mode:

- The conveyor runs automatically when the system is enabled
- Product detection sensors trigger the labelling cycle
- The PLC controls timing for label application and conveyor movement
- Each product is processed in a continuous sequence

This mode is used during normal production operations where consistent and uninterrupted output is required.

6.2 Manual Mode (Operator Control Mode)

Manual Mode allows individual control of system components for testing, setup, or controlled operation. In this mode, the PLC permits direct operator commands through the HMI.

In this mode:

- Conveyor motor can be started or stopped independently
- Label applicator can be activated manually
- Sensors can be monitored in real time
- Each subsystem can be tested separately

Manual Mode is typically used during commissioning, troubleshooting, or setup activities where controlled operation of individual components is required.

6.3 Maintenance Mode (Testing and Diagnostic Mode)

Maintenance Mode is used for system inspection, fault diagnosis, and preventive maintenance activities. This mode restricts full automation and allows safe interaction with system components.

In this mode:

- Individual sensors and actuators can be tested
- Conveyor operation is controlled at low or jog speed where applicable
- I/O signals from PLC can be monitored for diagnostics
- Fault conditions can be identified and verified

This mode ensures safe access to mechanical and electrical components while minimizing the risk of unintended system activation.

6.4 Mode Selection and Transition Control

Mode selection is performed through the HMI, and all transitions between modes are managed by PLC logic. The system ensures that mode switching occurs only under safe conditions, such as when the conveyor is stopped and no active process cycle is running. This prevents operational conflicts and ensures system stability during transitions.

7. Start-Up Procedure

The start-up procedure outlines the correct sequence for powering and initializing the auto labelling system to ensure safe and reliable operation. The following steps must be followed in the specified order before initiating any automatic or manual operation.

7.1 Power ON Sequence

Before energizing the system, it must be confirmed that all safety conditions are met, including the absence of mechanical obstructions and completion of pre-start safety checks.

The power ON sequence is as follows:

- Switch ON the main power supply to the control panel
- Energize the PLC power supply unit
- Activate power to the conveyor drive system
- Ensure auxiliary systems such as sensors and actuators receive stable power

Once power is applied, indicator lights on the control panel should confirm system energization.

7.2 PLC Initialization

After power is supplied, the PLC undergoes an initialization process where internal memory, input/output modules, and communication interfaces are activated.

During PLC initialization:

- The PLC performs a hardware self-check
- Input and output modules are scanned and registered
- Control logic is loaded and executed in cyclic scan mode
- Communication with field devices and HMI is established

The system should remain in a stopped or safe state until initialization is complete.

7.3 HMI Boot-Up

The HMI panel initializes after receiving stable communication from the PLC. It loads the operator interface, including system dashboards, control screens, and status indicators.

During this process:

- The HMI establishes communication with the PLC
- System screens and control menus are loaded
- Real-time data variables are synchronized

- Alarm history and status indicators become active
- Operators must verify that all displays are functioning correctly before proceeding.

7.4 System Readiness Check

Once both PLC and HMI are fully initialized, a system readiness check is performed to confirm that all subsystems are in a safe and operational state.

The readiness check includes verification of:

- No active fault or alarm conditions
- All sensors reporting valid signals
- Conveyor system in stopped state
- Labelling unit in home or idle position
- Emergency stop circuit in normal (non-activated) condition

When all conditions are satisfied, the system is considered ready for operation and can be switched to the desired operating mode through the HMI.

8. Normal Operation Procedure

The normal operation procedure describes the standard sequence of actions performed by the auto labelling system during automatic production mode. This sequence is fully controlled by the PLC and is executed based on real-time sensor feedback and programmed control logic.

8.1 Product Placement on Conveyor

Products are placed manually or supplied through an upstream process onto the conveyor system. The conveyor transports the product towards the labelling station in a controlled and continuous manner. Proper spacing between products is maintained to ensure accurate detection and labelling.

8.2 Product Detection

As the product moves along the conveyor, it passes through a detection zone where sensors continuously monitor its presence. When a product enters the sensing range, the sensor generates a digital input signal to the PLC indicating product presence and position.

The PLC evaluates this signal in conjunction with system timing logic to determine readiness for the labelling process.

8.3 PLC-Controlled Conveyor Coordination

Upon valid product detection, the PLC coordinates conveyor movement to ensure correct positioning at the labelling point. Depending on system configuration, the conveyor may slow down, pause momentarily, or continue at a synchronized speed to maintain accurate label placement timing.

This synchronization ensures that each product is correctly aligned under the labelling mechanism before activation.

8.4 Label Application Process

Once the product reaches the correct position and timing conditions are met, the PLC activates the label applicator unit. The labelling mechanism dispenses and applies the label onto the product surface.

The activation is precisely timed to ensure:

- Correct label alignment

- Consistent application position
- No overlap or misplacement

After completion of the labelling cycle, the applicator resets to its standby state for the next product.

8.5 Verification Step (If Applicable)

In systems equipped with verification mechanisms, a post-labelling check may be performed using sensors or vision systems. This step confirms whether the label has been correctly applied.

If an error is detected, the system may trigger a rejection signal or flag the product for manual inspection.

8.6 Product Exit from System

After successful labelling (and verification, if applicable), the conveyor continues to transport the product out of the labelling zone. The product then exits the system and moves to the next stage of the production or packaging line.

The PLC resets the cycle and prepares the system for the next incoming product, ensuring continuous operation without interruption.

9. HMI Operation Guide

The Human-Machine Interface (HMI) provides the primary user interface for monitoring and controlling the auto labelling system. It enables operators to interact with the PLC in real time through a graphical display, allowing safe and efficient system supervision. All commands entered via the HMI are transmitted to the PLC for execution, while system feedback is continuously displayed for operational awareness.

9.1 Start/Stop Controls

The HMI includes dedicated Start and Stop controls used to initiate or halt system operation.

Start Button: Initiates the system sequence in the selected operating mode, enabling conveyor movement and automated labelling logic execution.

Stop Button: Safely stops conveyor movement and disables the labelling cycle after completing the current operation sequence.

The PLC ensures that Stop commands are executed in a controlled manner to avoid abrupt system interruption.

9.2 Mode Selection (Auto / Manual)

The HMI allows operators to select the operating mode based on production or maintenance requirements.

Automatic Mode: Enables continuous, PLC-controlled operation of the full labelling cycle.

Manual Mode: Allows individual control of system components such as conveyor movement and label activation for testing or setup purposes.

Mode transitions are managed by PLC logic to ensure safe switching between operational states.

9.3 Conveyor Speed Control

The HMI provides control over conveyor speed settings where applicable. Operators can adjust speed parameters based on production requirements.

- Speed adjustments are transmitted to the PLC as setpoint values
- The PLC regulates motor drive output based on configured parameters
- Changes in speed are applied in a controlled manner to maintain system stability
- This function ensures flexibility in handling different product sizes and production rates.

9.4 Alarm Display and Reset

The HMI continuously monitors system status and displays active alarms or fault conditions detected by the PLC.

- Alarms may include sensor failure, conveyor jam, or communication faults
- Each alarm is displayed with a descriptive message and timestamp
- The reset function allows operators to acknowledge and clear alarms after resolving the issue
- The PLC prevents system restart until all critical alarms are cleared.

9.5 System Status Indicators

The HMI provides real-time visual indicators for system monitoring, including:

- Conveyor running/stopped status
- Active operating mode
- Sensor detection status
- Label application cycle status
- System readiness and fault indicators

These indicators help operators quickly assess system health and operational state, ensuring safe and efficient control of the labelling process.

10. Alarm & Fault Indications

The auto labelling system is equipped with alarm and fault detection functionality to ensure safe operation and early identification of abnormal conditions. Alarms are generated by the PLC based on input signals, timing logic, and system diagnostics. These alarms are displayed on the HMI to inform the operator of the fault type and required corrective action.

10.1 Sensor Failure

A sensor failure alarm is triggered when the PLC detects inconsistent or missing signals from product detection or positioning sensors.

This condition may occur due to:

- Faulty or damaged sensor hardware
- Misalignment of sensor position
- Obstruction in the sensing path
- Wiring or connection issues

When this alarm is active, the system may pause automatic operation to prevent incorrect labelling or product handling errors.

10.2 Label Misalignment

Label misalignment occurs when the label is not applied correctly on the product surface or deviates from the intended position.

- Possible causes include:
- Incorrect timing between conveyor movement and label activation
- Mechanical slippage in the labelling unit
- Improper calibration of label applicator
- Variations in product positioning on the conveyor

The system may flag affected products for inspection and halt automatic operation if repeated misalignment is detected.

10.3 Conveyor Jam

A conveyor jam alarm is generated when product movement is obstructed or when the conveyor motor is unable to operate as expected.

This condition may be caused by:

- Physical obstruction on the conveyor path
- Excess product accumulation or improper spacing
- Motor overload or drive failure
- Mechanical wear or belt misalignment

When a jam is detected, the PLC immediately stops conveyor operation to prevent equipment damage.

10.4 PLC Communication Error

A PLC communication error alarm is triggered when there is a failure in data exchange between the PLC and connected devices such as the HMI, sensors, or motor drives.

Possible reasons include:

- Loss of network or communication link
- Faulty communication cable or interface module
- Configuration mismatch between devices
- Temporary signal interruption

In this condition, the system may enter a safe stop state until communication is restored and verified.

11. Shutdown Procedure

The shutdown procedure defines the correct sequence for safely powering down the auto labelling system. Proper shutdown ensures protection of equipment, prevents data loss in the PLC, and maintains system integrity for the next operation cycle. The system should always be brought to a controlled stop before isolating power.

11.1 Safe Stop of Conveyor System

The first step in the shutdown process is to stop the conveyor system in a controlled manner.

The Stop command is initiated via the HMI or control panel

The PLC ensures that the conveyor completes any active cycle before stopping

Product flow is cleared from the labelling zone where possible

Conveyor motor is brought to a complete stop in a controlled state

This prevents sudden stoppage that could lead to product misplacement or mechanical stress.

11.2 Disabling of Labelling System

Once the conveyor has stopped, the labelling unit is deactivated to ensure safe system conditions.

The PLC disables output signals to the labelling actuator

No further label dispensing is permitted

The system is placed in an idle or safe state

Remaining products in process are either completed or cleared depending on system logic

This step ensures that no unintended labelling actions occur during shutdown.

11.3 Power Down of PLC and HMI

After system operations are safely halted, control devices can be powered down.

The HMI is shut down through its standard interface

The PLC is allowed to complete its final scan cycle before power removal

Control panel power supply is switched off following safe electrical procedures

Indicator lights confirm loss of system power

It is important that the PLC is not abruptly disconnected without proper shutdown sequence unless in emergency conditions.

11.4 Emergency Shutdown Conditions

Emergency shutdown is used only in critical situations where immediate system stoppage is required to prevent injury, equipment damage, or operational hazards.

Emergency shutdown may be activated in cases such as:

- Sudden mechanical failure
- Safety hazard to personnel
- Conveyor blockage causing system risk
- Electrical faults or abnormal system behavior
- When emergency shutdown is activated:
- All outputs are immediately deactivated by the PLC
- Conveyor and labelling operations stop instantly

The system enters a safe state until reset and inspection are completed

Restart after an emergency shutdown must only be performed after identifying and resolving the root cause of the fault.

12. Basic Troubleshooting

This section provides basic troubleshooting guidance for common operational issues in the auto labelling system. It is intended to assist operators and maintenance personnel in identifying and resolving typical faults without requiring advanced system intervention. All troubleshooting activities should be performed only after ensuring safe system conditions.

12.1 Conveyor Not Running

If the conveyor system does not start or stops unexpectedly, the following checks should be performed:

Possible causes:

- No power supply to motor drive or control panel
- PLC output signal not activated
- Emergency stop circuit engaged
- Motor overload or drive fault condition
- Faulty start command from HMI or control switch

Corrective actions:

- Verify main power supply and control panel indicators
- Check HMI for active alarms or interlocks
- Ensure emergency stop is released and reset
- Inspect motor drive status and reset if required
- Re-initiate start command after clearing faults

12.2 No Label Applied

If products pass through the system without receiving labels, the issue may be related to timing, sensor input, or actuator malfunction.

Possible causes:

- Label applicator not triggered by PLC
- Incorrect sensor detection or timing mismatch
- Empty label supply or mechanical blockage
- Output signal failure from PLC to actuator

Corrective actions:

- Check label supply and mechanical condition of applicator
- Verify sensor alignment and detection signal in HMI

- Inspect PLC output status for label trigger signal
- Recalibrate timing parameters if required

12.3 Sensor Not Detecting Product

If the system does not detect products on the conveyor, the following issues should be considered:

Possible causes:

- Misaligned or obstructed sensor
- Dirty or damaged sensor surface
- Wiring or communication fault
- Incorrect sensor configuration in PLC logic

Corrective actions:

- Clean and realign sensor position
- Inspect sensor wiring and connections
- Test sensor input status in HMI or PLC diagnostics
- Replace faulty sensor if necessary

12.4 System Restart Procedure

If the system requires a restart due to faults or abnormal operation, the following steps should be followed:

- Stop the system using HMI or control panel
- Clear all active alarms and fault conditions
- Ensure conveyor and labelling unit are in safe state
- Perform a controlled power cycle of the control system if required
- Restart PLC and HMI and allow full initialization
- Re-select appropriate operating mode before resuming operation

System restart should only be performed after confirming that the root cause of the issue has been addressed to prevent recurrence of the fault.

13. Notes on System Operation

This section provides important operational considerations that influence the performance, reliability, and accuracy of the auto labelling system. These points are intended to support proper system usage and help prevent common operational inefficiencies.

13.1 Conveyor Timing Dependency

The performance of the labelling process is highly dependent on the timing and synchronization of the conveyor system. Accurate label placement is achieved only when product movement, sensor detection, and label activation are properly coordinated.

Any variation in conveyor speed or inconsistent product spacing may affect the timing of label application. Therefore, conveyor speed settings must remain stable during automatic operation, and adjustments should be made only under controlled conditions.

13.2 Sensor Alignment Importance

Correct alignment of sensors is critical for reliable product detection and system synchronization. Sensors that are misaligned, obstructed, or improperly positioned may result in incorrect detection signals, leading to missed labels or false triggering.

Regular inspection of sensor positioning is recommended to ensure that detection zones are accurately maintained. Sensor surfaces should also be kept clean and free from dust or physical obstruction to maintain consistent performance.

13.3 PLC Scan Cycle Behavior

The PLC operates on a continuous scan cycle consisting of input reading, logic execution, and output updating. This cycle is repeated continuously at high speed, allowing real-time control of the system.

Understanding this behavior is important because all system responses, including sensor detection and actuator activation, are dependent on scan timing. Any delay in input signal processing or output response is governed by the PLC scan cycle, which ensures deterministic and predictable system operation under normal conditions.